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Chapter 2
Units and Measurement

Introduction

Measurement of any physical quantity involves comparison with a certain basic, arbitrarily chosen, internationally accepted reference standard called unit. The result of a measurement of a physical quantity is expressed by a number accompanied by a unit.

PHYSICAL QUANTITIES

The quantities, which can be measured directly or indirectly, are called physical quantities.

There are two types of physical quantities- Fundamental quantities (Base quantities) and Derives quantities.

The physical quantities which are independent of each other and cannot be expressed in terms of other physical quantities are called fundamental quantities.
Eg : Length, mass time

Physical quantities, which can be expressed in terms of fundamental quantities, are called derived quantities.
Eg: volume, velocity, force

Fundamental and derived units

The units for the fundamental or base quantities are called **fundamental or base units.**
The units of length, mass and time, ie metre, kilogram and second are taken as fundamental units.

The units of derived quantities are called **derived units.**
Eg: Volume is measured in metre ^{3} or m ^{3}.
Velocity is expressed in meter per second (m/s)

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Systems of Units

The CGS, the FPS (or British) and the MKS system were in use extensively till recently.

The base units for length, mass and time in these systems were as follows :

- CGS system - centimetre, gram and second
- FPS system - foot, pound and second
- MKS system - metre, kilogram and second

The International System of Units

in 1971 the General Conference on Weights and Measures developed an internationally accepted system of units for measurement with standard scheme of symbols, units and abbreviations.

This is the Système Internationale d' Unites (French for International System of Units), abbreviated as SI system.

Now SI system is used in scientific, technical , industrial and commercial work.

In SI system there are seven base units and two supplementary units.

BASE QUANTITY	BASE UNIT	SYMBOL
Length	metre	m
Mass	kilogram	kg
Time	second	s
Electric Current	ampere	A
Thermodynamic Temperature	kelvin	K
Amount of Substance	mole	mol
Luminous Intensity	candela	cd

SUPPLEMENTARY QUANTITY	SUPPLEMENTARY UNITS	SYMBOL
Plane Angle	radian	rad
Solid Angle	steradian	sr

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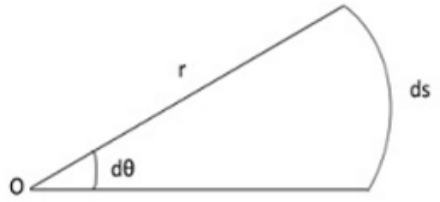
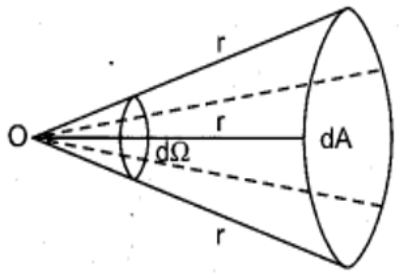


$$\text{Angle} = \frac{arc}{radius}$$

$$d\theta = \frac{ds}{r}$$



$$\text{Solid angle} = \frac{\text{Intercepted } Area}{Square\ of radius}$$

$$d\Omega = \frac{dA}{r^2}$$

Multiples and Sub multiples of units.

To express magnitude of physical quantities, which are very large or small, we use prefixes to the unit.

Submultiple	Prefix	Symbol	Multiple	Prefix	Symbol
10^{-1}	deci	d	10	deca	da
10^{-2}	centi	c	10^{2}	hecto	h
10^{-3}	milli	m	10^{3}	kilo	k
10^{-6}	micro	μ	10^{6}	mega	M
10^{-9}	nano	n	10^{9}	giga	G
10^{-12}	pico	p	10^{12}	tera	T
10^{-15}	femto	f	10^{15}	peta	P

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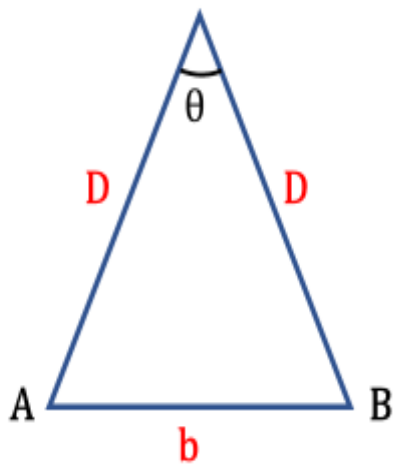
Measurement of Large Distances

Parallax method

Large distances such as the distance of a planet or a star from the earth can be measured by parallax method.

To measure the distance D of a far away planet S by the parallax method, we observe it from two different positions (observatories) A and B on the Earth, separated by distance AB = b.



$$\text{Angle} = \frac{arc}{radius}$$

$$\theta = \frac{b}{D}$$

$$\boxed{D = \frac{b}{\theta}}$$

D=distance of a planet from the earth
b=distance between the observatories
θ= parallax angle

Parallax method can also be used to determine the size or diameter of the planet or satellites.

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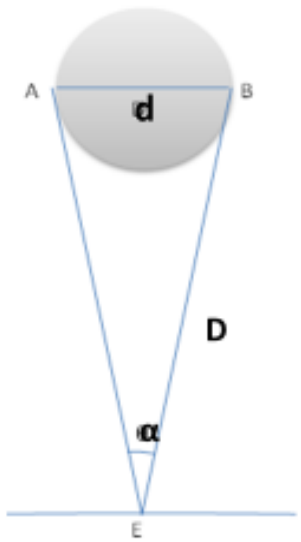

$$\text{Angle} = \frac{arc}{radius}$$

$$\alpha = \frac{d}{D}$$

$$d = \alpha D$$

d = the diameter planet.
α =the angle subtended by d at the earth
D = Distance to the planet from earth

Estimation of Very Small Distance

Optical microscope can resolve particles with sizes comparable with wavelength of light.

Electron microscopes can almost resolve atoms and molecules in a material.

Tunnelling microscopes can estimate the sizes of molecules.

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Range of Lengths

We also use certain special length units for short and large lengths.

1 fermi $= 1$ f $= 10^{-15}$ m
1 angstrom $= 1$ Å $= 10^{-10}$ m

1 astronomical unit $= 1$ AU (average distance of the Sun from the Earth)
$= 1.496 \times 10^{11}$ m

1 light year $= 1$ ly $= 9.46 \times 10^{15}$ m (distance that light travels with velocity of 3×10^{8} m s^{-1} in 1 year)

1 parsec $= 3.08 \times 10^{16}$ m (Parsec is the distance at which average radius of earth's orbit subtends an angle of 1 arc second)

Measurement Of Mass

While dealing with atoms and molecules **unified atomic mass unit (u),** is used.

1 unified atomic mass unit $= 1$u
$= (1/12)$ of the mass of an atom of carbon-12 isotope $^{12}_{6}C$ including the mass of electron

$$1u = 1.66 \times 10^{-27} \text{ kg}$$

Measurement Of Time

We now use an **atomic standard of time**, called **cesium clock**, sometimes called **atomic clock .** The cesium clock is based on the periodic vibrations produced in a cesium atom.

A cesium atomic clock is used at the National Physical Laboratory (NPL), New Delhi to maintain the Indian standard of time. The Indian Standard Time (IST) is linked to this atomic clock.

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Dimensions, Dimensional Formula and Dimensional Equation

The volume occupied by an object = length x breadth x height

The dimensions of volume represented as [V]

$$[V] = [L] \times [L] \times [L] = [L]^3 = [L^3].$$

$$[V] = [L^3]$$

$$[V] = [M^0 L^3 T^0]$$

Dimensions

Dimensions of a physical quantity are the powers (or exponents) to which the base quantities are raised to represent that quantity.

Volume has 0 Dimensions in mass, 3 Dimensions in length and 0 Dimensions in Time

Dimensional Formula

The expression which shows how and which of the base quantities represent the dimensions of a physical quantity is called the dimensional formula of the given physical quantity.

The dimensional formula of volume [V] is expressed as $[M^0 L^3 T^0]$

Dimensional Equation

An equation obtained by equating a physical quantity with its dimensional formula is called the dimensional equation of the physical quantity.

$$[V] = [M^0 L^3 T^0]$$

Dimensional Formula of some derived quantities

1. Area = Length x Breadth

$$[A] = [L] \times [L]$$

$$[A] = [L^2]. \quad \text{Or} \quad [A] = [M^0 L^2 T^0]$$

Unit of area = m^2

2. Volume = Length x Breadth x Height

$$[V] = [L] \times [L] \times [L]$$

$$[V] = [L^3]$$

Unit of volume = m^3

3. Density = $\frac{Mass}{Volume}$

$$[\rho] = \frac{[M]}{[L^3]}$$

$$[\rho] = [ML^{-3}].$$

Unit of density =kg m^{-3}

4 .Frequency = $\frac{1}{Time\ period}$

$$[f] = \frac{1}{[T]}$$

$$[f] = [T^{-1}]$$

Unit of frequency $= s^{-1}$

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5. $\text{Speed} = \frac{\text{Distance}}{\text{time}}$

$[s] = \frac{[L]}{[T]}$

$[s] = [LT^{-1}]$

Unit of speed $= m\ s^{-1}$

6. $\text{Velocity} = \frac{\text{Displacement}}{\text{time}}$

$[v] = \frac{[L]}{[T]}$

$[v] = [LT^{-1}]$

Unit of velocity $= m\ s^{-1}$

Speed and Velocity have same dimensional formula

7. Momentum = Mass x Velocity

$[p] = [M] \times [LT^{-1}]$

$[p] = [MLT^{-1}]$

Unit of momentum $= kg\ m\ s^{-1}$

8. Angular Momentum = momentum x Distance

$[L] = [MLT^{-1}] \times [L]$

$[L] = [M\ L^2\ T^{-1}]$

Unit of angular momentum $= kg\ m^2\ s^{-1}$

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9. $\text{Acceleration} = \frac{\text{Change in velocity}}{\text{time}}$

$[a] = \frac{[LT^{-1}]}{[T]}$

$[a] = [LT^{-1}] \times [T^{-1}]$

$[a] = [LT^{-2}]$

Unit of acceleration $= m\ s^{-2}$

10. Force = Mass x Acceleration

$[F] = [M] \times [LT^{-2}]$

$[F] = [MLT^{-2}]$

Unit of force $= kg\ m\ s^{-2}$ or newton(N)

$1 kgms^{-2} = 1N$

11. Impulse = Force x Time

$[I] = [MLT^{-2}] \times [T]$

$[I] = [MLT^{-1}]$

Unit of Impulse $= kg\ m\ s^{-1}$

Impulse and Momentum has the same dimension.

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12. Work = Force x Displacement

$[W] = [MLT^{-2}] \times [L]$

$[W] = [M L^2 T^{-2}]$

Unit of work $= kg\ m^2 s^{-2}$ or joule (J)

$1 kgm^2 s^{-2} = 1J$

13. Energy = Workdone

$[E] = [M L^2 T^{-2}]$

Unit of energy $= kg\ m^2 s^{-2}$ or joule (J)

14. Torque = Force x perpendicular Distance

$[\tau] = [MLT^{-2}] \times [L]$

$[\tau] = [M L^2 T^{-2}]$

Unit of torque $= kg\ m^2 s^{-2}$

Work, Energy and Torque have same dimensional formula

15. Pressure $= \frac{\text{Force}}{\text{Area}}$

$[P] = \frac{[MLT^{-2}]}{[L^2]}$

$[P] = [MLT^{-2}] \times [L^{-2}]$

$[P] = [ML^{-1}T^{-2}]$

Unit of pressure $= kg\ m^{-1} s^{-2}$ or pascal(Pa)

16. Stress $= \frac{\text{Force}}{\text{Area}}$

$[stress] = \frac{[MLT^{-2}]}{[L^2]}$

$[stress] = [ML^{-1}T^{-2}]$

Unit of stress $= kg\ m^{-1} s^{-2}$

Pressure and Stress have same dimensional formula

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17. $\text{Power} = \frac{\text{Work}}{\text{Time}}$

$$[P] = \frac{[M\,L^2\,T^{-2}]}{[T]}$$

$$[P] = [M\,L^2\,T^{-2}] \times [T^{-1}]$$

$$[P] = [M\,L^2\,T^{-3}]$$

Unit of power=$kg\,m^2\,s^{-3}$ or watt(W)

Physical quantities having no dimension and no unit

$$\text{Strain} = \frac{\text{Change in dimension}}{\text{Original dimension}} = \frac{[L]}{[L]} = [L^0]$$

$$\text{Relative Density} = \frac{\text{Density of substance}}{\text{Density of water}} = \frac{[ML^{-3}]}{[ML^{-3}]} = [L^0]$$

Physical quantities having units, but no dimension

Plane angle
Solid Angle
Angular Displacement

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Find the dimensional formula of Gravitational constant using equation $F = \frac{Gm_1m_2}{r^2}$

$$G = \frac{Fr^2}{m_1m_2}$$

$$[G] = \frac{[F][r^2]}{[m_1]\,[m_2]}$$

$$[G] = \frac{[MLT^{-2}][L^2]}{[M][M]}$$

$$= [MLT^{-2}]\,[L^2][M^{-1}][M^{-1}]$$

$$[G] = [M^{-1}L^3T^{-2}]$$

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Find the dimensional formula of Planck's constant using equation $\lambda = \frac{h}{mv}$

$$h = \lambda m v$$

$$[h] = [\lambda][m][v]$$

$$[h] = [L][M]\,[LT^{-1}]$$

$$[h] = [ML^2T^{-1}]$$

Find the dimensional formula of Coefficient of Viscosity using equation $F = \eta A\left(\frac{dv}{dx}\right)$

$$\eta = \frac{F}{A\left(\frac{dv}{dx}\right)}$$

$$\eta = \frac{F dx}{A\, dv}$$

$$[\eta] = \frac{[MLT^{-2}] \times [L]}{[L^2]\,[L\,T^{-1}]}$$

$$[\eta] = [MLT^{-2}] \times [L] \times [L^{-2}] \times [L^{-1}] \times [T]$$

$$[\eta] = [ML^{-1}T^{-1}]$$

Dimensional analysis and its applications

1) Checking the Dimensional Consistency(correctness) of Equations
2) Deducing Relation among the Physical Quantities

1) Checking the Dimensional Consistency(correctness) of Equations

The principle called the principle of homogeneity of dimensions is used to check the dimensional correctness of an equation.

The principle of homogeneity states that, for an equation to be correct, the dimesions of each terms on both sides of the equation must be the same.

The magnitudes of physical quantities may be added or subtracted only if they have the same dimensions.

If $X + Y = Z$
$[X] = [Y] = [Z]$

We cannot add 5m and 10kg
Velocity cannot be added to force.

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eg1) Check the dimensional consistency(correctness) or homogeneity of the equation

$$s = ut + \frac{1}{2}at$$

s= displacement
u =initial velocity
a=acceleration
t=time

$$[s] = L$$
$$[ut] = LT^{-1} \times T$$
$$= L$$
$$[\frac{1}{2}at] = LT^{-2} \times T$$
$$= LT^{-1}$$

Since the dimensions of all terms of the equation are not same, this equation is wrong.

2) Check the dimensional consistency(correctness) or homogeneity of the equation

$$s = ut + \frac{1}{2}at^2$$

s= displacement
u =initial velocity
a=acceleration
t=time

$$[s] = L$$
$$[ut] = LT^{-1} \times T$$
$$= L$$
$$[\frac{1}{2}at^2] = LT^{-2} \times T^2$$
$$= L$$

Since each term on both sides of equation has the same dimension, this equation is dimensionally correct

If an equation passes this consistency test it is not proved right.

3)Check the dimensional consistency(correctness) or homogeneity of the equation

$$s = ut + \frac{2}{3}at^2$$

s= displacement
u =initial velocity
a=acceleration
t=time

$$[s] = L$$
$$[ut] = LT^{-1} \times T$$
$$= L$$
$$[\frac{2}{3}at^2] = LT^{-2} \times T^2$$
$$= L$$

Since each term on both sides of equation has the same dimension, this equation is dimensionally correct.

Eventhough this equation is is dimensionally correct,it is not an exact equation.

A dimensionally correct equation need not be an exact (correct) equation, but a dimensionally wrong (incorrect) must be wrong.

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4) Check the dimensional consistency(correctness) or homogeneity of the equation

$$\frac{1}{2}mv^2 = mgh$$

m=mas of th body
v= velocity of body
g=acceleration due to gravity
h=height

$$[\tfrac{1}{2}mv^2] = M\ [LT^{-1}]^2$$
$$= ML^2T^{-2}$$

$$[mgh] = M\ LT^{-2}\ L$$
$$= ML^2T^{-2}$$

The dimensions of LHS and RHS are the same and hence the equation is dimensionally correct.

5) Check the dimensional correctness of the equation $E = m\,c^2$

$$[E] = ML^2T^{-2}$$

E=energy
m=mass
c=velocity of light

$$[m\,c^2] = M\ [LT^{-1}]^2$$
$$= ML^2T^{-2}$$

The dimensions of LHS and RHS are the same and hence the equation is dimensionally correct.

6) In the given equation $v = x + at$, find the dimensions of x.
(where v = velocity, a= acceleration, t = time)

$v = x + at$
$[v] = [x] = [at]$
$[x] = [v]$
$[x] = LT^{-1}$

7)In the given equation $x = a + bt + ct^2$, find the dimensions of a,b and c.
(where x is in metres and t in seconds)

$x = a + bt + ct^2$
$[x] = [a] = [bt] = [ct^2]$

$[a] = [x]$
$[a] = L$

$[bt] = [x]$
$[b] \times T = L$
$[b] = \frac{L}{T}$
$[b] = LT^{-1}$

$[ct^2] = [x]$
$[c] \times T^2 = L$
$[c] = \frac{L}{T^2}$
$[c] = LT^{-2}$

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8) The SI unit of energy is J = kg m^2 s^{-2}; that of speed v is ms^{-1} and of acceleration a is ms^{-2} Using dimensional arguments, find which of the formulae can be considered for kinetic energy (K)

(a) K= m^2v^3

(b) $K = \frac{1}{2}mv^2$

(c) K=ma

(d) $K = \frac{3}{16}mv^2$

(e) $K = \frac{1}{2}mv^2 + ma$

(a) $K = m^2v^3$

$$M L^2 T^{-2} = [M]^2 [LT^{-1}]^3$$

$$\neq M^2 L^3T^{-3}$$

(b) $K = \frac{1}{2}mv^2$

$$M L^2 T^{-2} = [M] [LT^{-1}]^2$$

$$= ML^2T^{-2}$$

(c) $K = ma$

$$M L^2 T^{-2} = M [LT^{-2}]$$

$$\neq MLT^{-2}$$

(d) $K = \frac{3}{16}mv^2$

$$M L^2 T^{-2} = [M] [LT^{-1}]^2$$

$$= ML^2T^{-2}$$

(e) $K = \frac{1}{2}mv^2 + ma$

$$[M L^2 T^{-2}] = [M] [LT^{-1}]^2 + [M] [LT^{-2}]$$

$$\neq ML^2T^{-2} + MLT^{-2}$$

Since dimensions of all terms are the same for Equations (b) and (d) , these equations can be considered as the equation for kinetic energy.

9.The Van der waals equation of 'n' moles of a real gas is $(P+\frac{a}{V^2})(V-b)=nRT$. Where P is the pressure, V is the volume, T is absolute temperature, R is molar gas constant and a, b, c are Van der waal constants. Find the dimensional formula for a and b.

$$(P+\frac{a}{V^2})(V-b)=nRT.$$

By principle of homegeneity, the quantities with same dimensions can be added or subtracted.

$$[P] = [\frac{a}{V^2}]$$

$$[a] = [PV^2]$$

$$= ML^{-1}T^{-2} \times L^6$$

$$[a] = ML^5T^{-2}$$

$$[b] = [V]$$

$$[b] = L^3$$

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2) Deducing Relation among the Physical Quantities

We can deduce relation of a physical quantity which depends upto three physical quantities.

1) Derive the equation for kinetic energy(E) of a body of mass m moving with velocity v.

$E \alpha\ m^x\ v^y$

$E = k\, m^x v^y \longrightarrow (1)$

Writing the dimensions on both sides,

$M\, L^2\, T^{-2} = M^x\, (LT^{-1})^y$

$M^1 L^2\ T^{-2} = M^x\ L^y\ T^{-y}$

equating the dimensions on both sides,

$x = 1$
$y = 2$

Substituting in eq (1)

$E = k\, m^1\, v^2$

$$\boxed{E = k\, mv^2}$$

2) Suppose that the period of oscillation of the simple pendulum depends on its mass of the bob (m), length (l) and acceleration due to gravity (g). Derive the expression for its time period using method of dimensions.

$T \propto m^x\ l^y\ g^z$

$T = k\, m^x\, l^y\, g^z \longrightarrow (1)$

Writing the dimensions on both sides,

$M^0 L^0 T^1 = M^x\ L^y\ (LT^{-2})^z$

$M^0 L^0 T^1 = M^x\ L^y\ L^z\ T^{-2z}$

$M^0 L^0 T^1 = M^x\ L^{y+z}\ T^{-2z}$

equating the dimensions on both sides,

$x = 0$

$y + z = 0$

$-2z = 1 \qquad z = \frac{-1}{2}$

$y + \frac{-1}{2} = 0 \qquad y = \frac{1}{2}$

$$\boxed{T = km^0\, l^{1/2} g^{-1/2}}$$

$T = k \frac{l^{1/2}}{g^{1/2}}$

$T = k \frac{\sqrt{l}}{\sqrt{g}}$

$$\boxed{T = k \sqrt{\frac{l}{g}}}$$

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Limitations of Dimensional Analysis

1) Dimensional analysis check only the dimensional correctness of an equation, but not the exact correctness.
2) The dimensionless constants cannot be obtained by this method.
3) We cannot deduce a relation, if a physical quantity depends on more than three physical quantities.
4) The method cannot be considered to derive equations involving more than one term
5) A formula containing trigonometric, exponential and logarithmic function can not be derived from it.
6) It does not distinguish between the physical quantities having same dimensions.

Accuracy , Precision of Instruments And Errors In Measurement

The result of every measurement by any measuring instrument contains some uncertainty. This uncertainty is called error.

Every calculated quantity which is based on measured values, also has an error.

Accuracy and Precision.

Accuracy:-The accuracy of a measurement is a measure of how close the measured value is to the true value of the quantity.

Precision :-Precision tells us to what resolution or limit the quantity is measured.

A measurement can have more accuracy with less precision or less accuracy with more precision

Errors In Measurement

Every measurement is approximate due to errors in measurement.

Errors

Systematic errors Random errors.

Systematic errors

The systematic errors are those errors that tend to be in one direction, either positive or negative.

Random errors

The random errors are those errors, which occur irregularly and hence are random with respect to sign and size.

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Some of the sources of systematic errors are :

a)Instrumental errors that arise from the errors due to imperfect design or calibration of the measuring instrument, zero error in the instrument, etc.

(b) Imperfection in experimental technique or procedure

(c)Personal errors that arise due to an individual's bias, lack of proper setting of the apparatus or individual's carelessness

Sources of Random errors

These can arise due to random and unpredictable fluctuations in experimental conditions (e.g. unpredictable fluctuations in temperature, voltage supply, mechanical vibrations of experimental set-ups, etc), personal (unbiased) errors by the observer taking readings, etc

Least count error

The least count error is the error associated with the resolution of the instrument.
The smallest value that can be measured by the measuring instrument is called its least count.

For example, A metre scale has a least count of 0.1cm.
A Vernier Callipers has the least count as 0.01cm.
A Screw guage and Spherometer may have a least count of 0.001 cm.

Using instruments of higher precision, improving experimental techniques, etc., we can reduce the least count error

Absolute Error, Relative Error and Percentage Error

True value or Mean value or Average value

Suppose the values obtained in several measurements are $a_1, a_2, a_3, \dots \dots, a_n$
The arithmetic mean of these values is taken as the True value.

$$a_{mean} = \frac{a_1 + a_2 + \cdots + a_n}{n}$$

$$a_{mean} = \frac{1}{n}\sum_{i=1}^{n} a_i$$

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Absolute Error

The magnitude of the difference between the true value of the quantity and the individual measurement value is called the absolute error of the measurement.

$$\Delta a_1 = |a_1 - a_{mean}|$$

$$\Delta a_2 = |a_2 - a_{mean}|$$

.....

.....

$$\Delta a_n = |a_n - a_{mean}|$$

Mean Absolute Error

The arithmetic mean of all the absolute errors is taken as the final or mean absolute error.

$$\boldsymbol{\Delta a_{mean} = \frac{\Delta a_1 + \Delta a_2 + \cdots + \Delta a_n}{n}}$$

$$\Delta a_{mean} = \frac{1}{n}\sum_{i=1}^{n} \Delta a_i$$

Relative Error

The relative error is the ratio of the mean absolute error $\boldsymbol{\Delta a_{mean}}$ to the mean value $\boldsymbol{a_{mean}}$ of the quantity measured.

$$\delta a = \frac{\Delta a_{mean}}{a_{mean}}$$

Percentage Error

Relative error expressed in percent is called Percentage Error.

$$\text{Percentage Error} = \frac{\Delta a_{mean}}{a_{mean}} \; x \; 100\%$$

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Problem

We measure the period of oscillation of a simple pendulum. In successive measurements, the readings turn out to be 2.63 s, 2.56 s, 2.42 s, 2.71s and 2.80 s. Calculate the absolute errors, relative error and percentage error.

True value $a_{mean} = \frac{a_1+a_2+\cdots+a_n}{n}$

$$a_{mean} = \frac{2.63+2.56+2.42+2.71+2.80}{5}$$

$$= \frac{13.12}{5}$$

$$= 2.624 \text{ s}$$

$$\boldsymbol{a_{mean} = 2.62 \text{ s}}$$

(As the periods are measured to a resolution of 0.01 s, all times are to the second decimal; it is proper to put this mean period also to the second decimal.)

The absolute errors in the measurements are

$$\Delta a_1 = |2.63 \text{ s} - 2.62 \text{ s}| = 0.01 \text{ s}$$

$$\Delta a_2 = |2.56 \text{ s} - 2.62 \text{ s}| = 0.06 \text{ s}$$

$$\Delta a_3 = |2.42 \text{ s} - 2.62 \text{ s}| = 0.20 \text{ s}$$

$$\Delta a_4 = |2.71 \text{ s} - 2.62 \text{ s}| = 0.09 \text{ s}$$

$$\Delta a_5 = |2.80 \text{ s} - 2.62 \text{ s}| = 0.18 \text{ s}$$

Mean absolute error , $\Delta a_{mean} = \frac{\Delta a_1+\Delta a_2+\cdots+\Delta a_n}{n}$

$$\Delta a_{mean} = \frac{(0.01+0.06+0.20+0.09+0.18)\text{s}}{5}$$

$$= \frac{0.54\ s}{5}$$

$$\boldsymbol{\Delta a_{mean} = 0.11 \text{ s}}$$

Relative Error, $\delta a = \frac{\Delta a_{mean}}{a_{mean}}$

$$= \frac{0.11\ s}{2.6}$$

$$\boldsymbol{\delta a = 0.04}$$

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(As the arithmetic mean of all the absolute errors is 0.11 s, there is already an error in the tenth of a second.So second decimal digit 6 can be avoided.)

$$\text{Percentage Error} = 0.04\ x\ 100\% = 4\%$$

Combination of Errors

(a)Error of a sum or a difference

Rule 1: When two quantities are added or subtracted, the absolute error in the final result is the sum of the absolute errors in the individual quantities.

Suppose two physical quantities A and B have measured values $A \pm \Delta A$, $B \pm \Delta B$ respectively, where ΔA and ΔB are their absolute errors.

Then the error ΔZ in the sum $Z = A + B$ is given by

$$\Delta Z = \Delta A + \Delta B$$

For the difference $Z = A - B$ also

$$\Delta Z = \Delta A + \Delta B$$

Problem1

Two resistors of resistances R_1=(8±0.4) ohm and R_2=(6±0.3) ohm are connected in series, determine the effective resistance.

$$R = R_1 + R_2 = 8+6 = 14 \text{ ohm}$$

Error in measurement is,

$$\Delta R = \Delta R_1 + \Delta R_2 = 0.4+0.3 = 0.7 \text{ ohm}$$

Effective resistance R=14±0.7 ohm

Problem2

The temperatures of two bodies measured by a thermometer are $T_1 = (50 \pm 0.5)^0$C and $T_2 = (20 \pm 0.5)^0$C. Calculate the temperature difference and the error therein.

$$T = 50-20 = 30^0\ C$$

Error in measurement is,

$$\Delta T = \Delta T_1 - \Delta T_2 = 0.5+0.5 = 1^0 C$$

Temperature difference is, $T = (30 \pm 1)^0$C.

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(b) Error of a product or a quotient

Rule 2 : When two quantities are multiplied or divided, the relative error in the result is the sum of the relative errors in the multipliers.

Suppose Z = AB and the measured values of A and B are A ± ΔA and B ± ΔB.

The maximum possible relative error in Z is

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

For Z $=\frac{A}{B}$ also

$$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Problem3

The resistance R$=\frac{V}{I}$ where V = (100±5)V and I = (10±0.2)A. Find the relative error percentage error in R.

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\frac{\Delta R}{R} = \frac{5}{100} + \frac{0.2}{10}$$

$$\frac{\Delta R}{R} = 0.05 + 0.02 = 0.07$$

Percentage error $=\frac{\Delta R}{R}\ x\ 100\%$

=0.07 x 100 =7%

Problem4

The percentage error in the measurement of length and breadth of a rectangle are 4% and 1.5% respectively. Calculate the percentage error in the measurement of area of the rectangle?

A=l×b

Maximum possible relative error is

$$\frac{\Delta A}{A} = \frac{\Delta l}{l} + \frac{\Delta b}{b}$$

$$\frac{\Delta A}{A}\ x\ 100\% = \frac{\Delta l}{l}\ x\ 100\% + \frac{\Delta b}{b}\ x\ 100\%$$

=4% + 1.5%

=5.5%

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(c) Error in case of a measured quantity raised to a power

Rule 3 : The relative error in a physical quantity raised to the power 'n' is the 'n' times the relative error in the individual quantity.

If $Z = A^n$

$$\frac{\Delta Z}{Z} = n\frac{\Delta A}{A}$$

If $Z = A^3$

$$\frac{\Delta Z}{Z} = 3\,x\,\frac{\Delta A}{A}$$

Problem5

The percentage error in the measurement of radius of a sphere is 2%. Calculate the percentage error in the measurement of surface area and volume of the sphere?

$$\frac{\Delta r}{r}\,x\ 100\% = 2\%;$$

The surface area of sphere is given by $S = 4\pi r^2$

Hence percentage error in S,

$$\frac{\Delta r}{r}\,x\ 100\% = 2\times\frac{\Delta r}{r}\,x\ 100\%$$

$$= 2\times 2\% = 4\%$$

The Volume of sphere is given by $V = \frac{4}{3}\pi r^3$

Hence percentage error in V,

$$\frac{\Delta V}{V}\,x\ 100\% = 3\times\frac{\Delta r}{r}\,x\ 100\%$$

$$= 3\times 2\% = 6\%$$

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Problem 6

A physical quantity P is related to four observables a, b, c and d as, $P=\frac{a^3b^2}{\sqrt{c}\times d}$. The percentage errors of measurement in a, b, c and d are 1%, 3%, 4% and 2%, respectively. What is the percentage error in the quantity P?.

Ans

We have $P=\frac{a^3b^2}{\sqrt{c}\times d}$,therefore percentage error in P is

$$\frac{\Delta P}{P}\times 100\%=3\times\frac{\Delta a}{a}\times 100\%+2\times\frac{\Delta b}{b}\times 100\%+\frac{1}{2}\times\frac{\Delta c}{c}\times 100\%+\frac{\Delta d}{d}\times 100\%$$

i.e, $\frac{\Delta P}{P}\times 100\%=3\times 1\%+2\times 3\%+\frac{1}{2}\times 4\%+2\%=13\%$

Problem 7

Example 2.12 The period of oscillation of a simple pendulum is $T=2\pi\sqrt{L/g}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90 s using a wrist watch of 1 s resolution. What is the accuracy in the determination of g ?

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T^2 = 4\pi^2\frac{l}{g}$$

$$g = 4\pi^2\frac{l}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + 2\frac{\Delta T}{T}$$

$$\frac{\Delta g}{g} = \frac{0.1}{20} + 2\,x\,\frac{1}{90} = 0.032$$

$$\frac{\Delta g}{g}x\ 100\% = \frac{0.1}{20}x\ 100x\ 100\% + 2\,x\,\frac{1}{90}x\ 100\%$$

$$= 3\ .2\%$$

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Significant Figures

The result of measurement is a number that includes all digits in the number that are known reliable plus the first digit that is uncertain.

The reliable digits plus the first uncertain digit in a measurement are known as significant digits or significant figures.

If the period of oscillation of a simple pendulum is 1.62 s, the digits 1 and 6 are reliable and certain, while the digit 2 is uncertain

Rules for the determination of number of significant figures

A choice of change of different units does not change the number of significant digits or figures in a measurement.

For example, the length 2.308 cm has four significant figures.

But in different units, the same value can be written
0.02308 m - 4 significant figures.

23.08 mm - 4 significant figures.

23080 μm - 4 significant figures.

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Rule1: All the non-zero digits are significant.

38 - 2

123 - 3

23.453 - 5

Rule2: All the zeros between two non-zero digits are significant, no matter where the decimal point is,

1204 - 4

30007 - 5

20.03 - 4

Rule3: If the number is less than one, the zero(s) on the right of decimal point but to the left of the first non-zero digit are not significant. The zero conventionally put on the left of decimal point for a number less than one is not significant.

0.032 - 2

0.004002 - 4

0.0132 - 3

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Rule4:For a number without any decimal point the terminal or trailing zero(s) are not significant.

4200 - 2

23040 - 4

38100 - 3

Rule5: For a number with a decimal point, the trailing zero(s) are significant.

20.0 - 3

43.00 - 4

1203.0 - 5

Rule 6:The power of 10, in scientific notation is irrelevant to the determination of significant figures.

Now suppose we change units,

then 4.700 m = 470.0 cm = 4700 mm = 0.004700 km

All these measurements have four significant figures and a mere change of units cannot change the number of significant figures.

To remove such ambiguities in determining the number of significant figures, the best way is to report every measurement in scientific notation (in the power of 10).

$$4.700 \text{ m}$$

$$= 4.700 \times 10^2 \text{ cm}$$

$$= 4.700 \times 10^3 \text{ mm}$$

$$= 4.700 \times 10^{-3} \text{ km}$$

All these numbers have 4 significant namaste figures.

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Rounding off the Uncertain Digits

1) If the insignificant digit to be dropped is more than 5, the preceding digit is raised by 1

A number 2.746 rounded off to three significant figures is 2.75

Here the insignificant digit , 6 > 5 and hence 1 is added to the preceeding digit 4 .(4+1=5)

2) If the insignificant digit to be dropped less than 5, the preceding digit is left unchanged .

A number 2.743 rounded off to three significant figures is be 2.74.

Here the insignificant digit , 3< 5 and hence the preceeding number 4 does not change.

3) **If the insignificant digit to be dropped is 5,**

Case i) If the preceding digit is even, the insignificant digit is simply dropped.

A number 2.745 rounded off to three significant figures is 2.74.

Here the preceding digit 4 is even and hence 5 is simply dropped.

Case ii-) If the preceding digit is odd, the preceding digit is raised by 1.

A number 2.735 rounded off to three significant figures is 2.74

Here the preceding digit 3 ,is odd and hence 1 is added to It. (3+1=5)

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Rules for Arithmetic Operations with Significant Figures

(1)In multiplication or division, the final result should retain as many significant figures as are there in the original number with the least significant figures.

Eg:If mass of an object is measured to be, 4.237 g (four significant figures) and its volume is measured to be 2.51cm3(3 significant figures), then find its density in appropriate significant figures.

$$\text{Density} = \frac{mass}{volume} = \frac{4.237\ g}{2.51 cm^3} = 1.688047$$

As per rule the final result should be rounded to 3 significant figures .

So the answer is 1.69 g/ cm^3

(2) In addition or subtraction, the final result should retain as many decimal places as are there in the number with the least decimal places.

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Eg:Find the sum of the numbers 436.32 g, 227.2 g and 0.301 g to appropriate significant figures.

436.32 g +		(2 decimal places)
227.2 g +		(1 decimal place)
0.301 g		(3 decimal places)
663.821 g		

As per rule ,the final result should be rounded to 1 decimal place.

So the answer 663.8 g

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